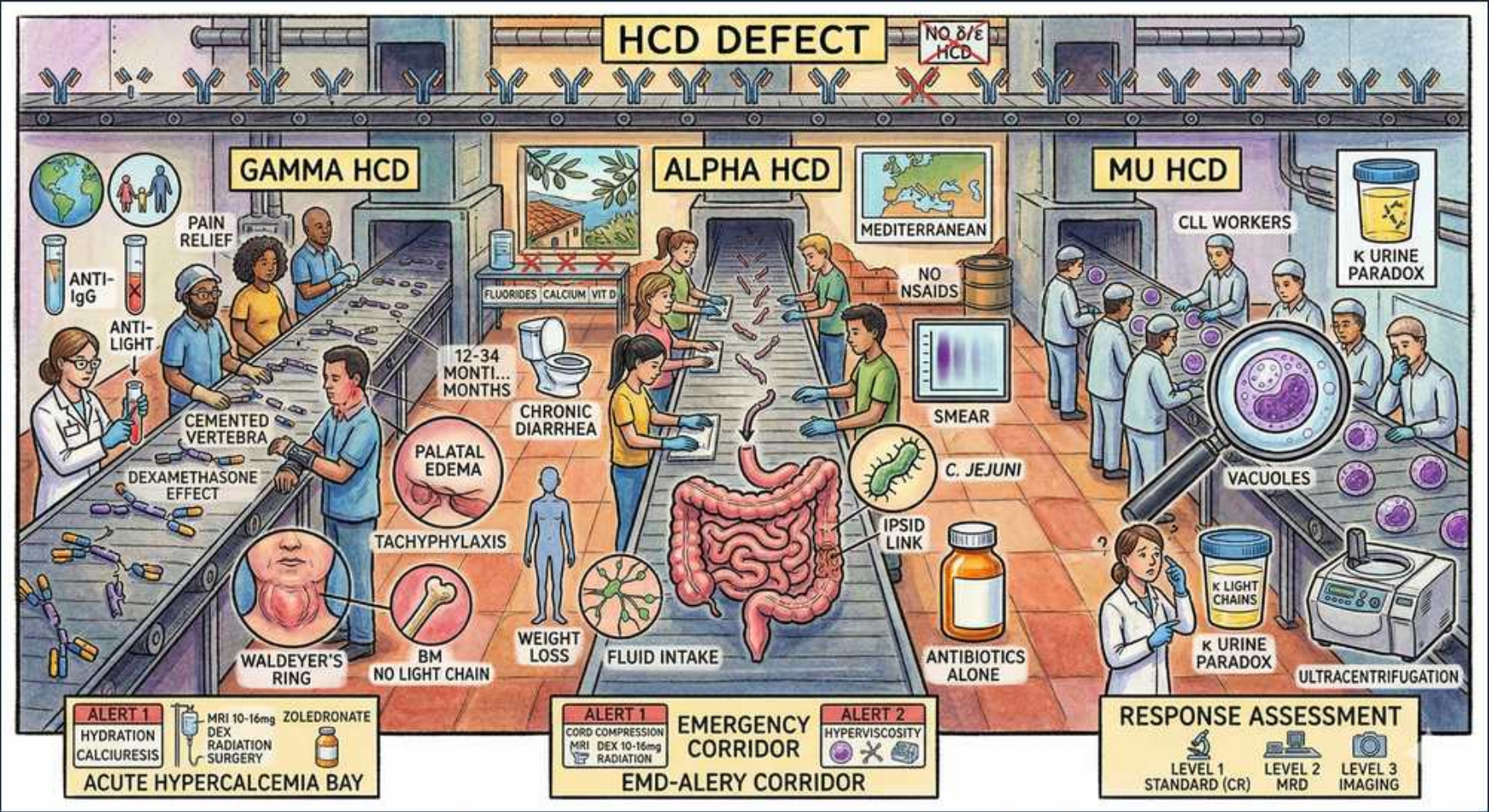


Heavy Chain Diseases — Gamma, Alpha & Mu



1. Heavy-chain defect (no light chain)

WHAT YOU SEE

Factory banner: 'HCD defect — no light chain.'

WHAT IT ENCODES

Heavy-chain diseases produce a truncated, incomplete immunoglobulin heavy chain WITHOUT an associated light chain. This causes the characteristic abnormal/odd immunofixation pattern.

BOARD TRAP

HCDs make a heavy chain alone — so the FLC ratio and light-chain studies are paradoxically normal/absent. That's the diagnostic fingerprint.

2. Gamma HCD (Franklin disease)

WHAT YOU SEE

Gamma assembly line, anti-IgG/anti-light dissociation.

WHAT IT ENCODES

Gamma HCD is a lymphoplasmacytic disorder resembling lymphoma. Presents with lymphadenopathy, hepatosplenomegaly, constitutional symptoms, and notably palatal/uvular edema (Waldeyer's ring involvement).

BOARD TRAP

Palatal edema + waxing-waning lymphadenopathy points to GAMMA HCD (Franklin), not WM or ordinary lymphoma.

3. Palatal edema (Waldeyer's ring)

WHAT YOU SEE

Throat with swollen palate and Waldeyer's ring.

WHAT IT ENCODES

Edema of the soft palate/uvula from involvement of Waldeyer's ring lymphoid tissue is a hallmark physical finding of gamma HCD.

BOARD TRAP

Uvular/palatal edema is the buzz-phrase clue for gamma HCD — don't attribute it to angioedema.

4. Alpha HCD (IPSID / Seligmann)

WHAT YOU SEE

Alpha line: Mediterranean, GI corridor.

WHAT IT ENCODES

Alpha HCD is the most common HCD: immunoproliferative small intestinal disease (IPSID). Young Mediterranean/Middle-Eastern patients with chronic diarrhea, malabsorption, weight loss; small-bowel MALT lymphoma of IgA lineage.

BOARD TRAP

Mediterranean young adult + chronic diarrhea/malabsorption = ALPHA HCD (IPSID), not celiac or tropical sprue.

5. Alpha HCD — Campylobacter jejuni link

WHAT YOU SEE

C. jejuni microbe near the gut.

WHAT IT ENCODES

Early IPSID is associated with Campylobacter jejuni infection and can RESPOND TO ANTIBIOTICS at the early (lymphoplasmacytic) stage before transforming to aggressive lymphoma.

BOARD TRAP

Early alpha HCD/IPSID may be cured with antibiotics alone (C. jejuni) — a unique, testable feature; advanced disease needs chemo.

6. Mu HCD

WHAT YOU SEE

Mu line: CLL workers, vacuolated cells.

WHAT IT ENCODES

Mu HCD is the rarest, associated with chronic lymphocytic leukemia / lymphoproliferative disease. Marrow shows characteristic VACUOLATED plasma cells; often hepatosplenomegaly without prominent lymphadenopathy.

BOARD TRAP

Vacuolated plasma cells + CLL-like picture = MU HCD; the vacuoles are the morphologic giveaway.

7. Mu HCD — kappa light chain in urine (paradox)

WHAT YOU SEE

Note: kappa urine light chain paradox.

WHAT IT ENCODES

Despite being a heavy-chain disease, mu HCD often shows free kappa light chains in the URINE — a recognized paradox because the heavy chain itself isn't secreted intact.

BOARD TRAP

Mu HCD can have urinary kappa light chains even though it's an HCD — don't let urinary light chain rule out HCD.

8. Avoid NSAIDs (alpha HCD GI)

WHAT YOU SEE

Sign: 'No NSAIDs.'

WHAT IT ENCODES

In alpha HCD with small-bowel involvement, NSAIDs are avoided to prevent GI bleeding/ulceration in already-diseased bowel; supportive nutrition and antibiotics are emphasized early.

BOARD TRAP

NSAID avoidance is a supportive-care point for alpha HCD GI disease — easy to overlook.

9. Immunofixation / no light chain on smear

WHAT YOU SEE

Smear and serum studies.

WHAT IT ENCODES

Diagnosis rests on serum/urine immunofixation showing a heavy chain reacting with anti-heavy-chain antiserum but NOT with anti-light-chain — confirming free heavy chain.

BOARD TRAP

Standard SPEP may miss HCD; immunofixation with anti-heavy/anti-light antisera is required to catch the dissociated heavy chain.

10. Ultracentrifugation / FLC paradox

WHAT YOU SEE

Ultracentrifuge + FLC paradox note.

WHAT IT ENCODES

Free light-chain assays are typically normal because no clonal light chain is overproduced; specialized testing (ultracentrifugation, immunofixation) identifies the abnormal heavy chain.

BOARD TRAP

A normal FLC ratio does NOT exclude HCD — the abnormality is in the heavy chain, not the light chain.

11. Acute hypercalcemia management

WHAT YOU SEE

Hypercalcemia bay: hydration, calcitonin, bisphosphonate.

WHAT IT ENCODES

When HCD/plasma-cell disorders cause hypercalcemia, treat with aggressive IV fluids, calcitonin for rapid effect, and zoledronate/bisphosphonate for durable control; steroids and dialysis if refractory.

BOARD TRAP

Bisphosphonates lower calcium over days; calcitonin gives the rapid early drop — sequence both, don't rely on one.

12. Cord-compression / EMD-ALERT corridor

WHAT YOU SEE

Emergency corridor: cord compression, dexamethasone, radiation.

WHAT IT ENCODES

Extramedullary disease and vertebral lesions can cause spinal cord compression — an oncologic emergency needing urgent steroids (dexamethasone) plus radiation or surgical decompression.

BOARD TRAP

Suspected cord compression: give dexamethasone immediately and get emergent MRI/RT — don't wait for biopsy.

13. Tachyphylaxis to calcitonin

WHAT YOU SEE

Tachyphylaxis sign in the hypercalcemia bay.

WHAT IT ENCODES

Calcitonin works fast but tachyphylaxis develops within ~48 hours, so it only bridges until bisphosphonate (zoledronate) takes effect.

BOARD TRAP

Calcitonin loses effect in 1-2 days (tachyphylaxis) — it's a bridge, not a definitive hypercalcemia therapy.

14. Response assessment levels (CR/MRD/imaging)

WHAT YOU SEE

Response assessment lab: Level 1 CR, MRD, imaging.

WHAT IT ENCODES

Response is graded from standard complete response (Level 1) to minimal residual disease (Level 2 MRD) to imaging-based response (Level 3), increasing in sensitivity.

BOARD TRAP

MRD negativity is deeper than conventional CR — a patient in 'CR' can still be MRD-positive.

15. Cemented vertebroplasty / bone support

WHAT YOU SEE

Cemented vertebra, dexamethasone effect.

WHAT IT ENCODES

For painful vertebral lesions, vertebroplasty/kyphoplasty and bisphosphonates support the skeleton while systemic therapy treats the clone.

BOARD TRAP

Vertebroplasty is supportive pain control — it does not treat underlying clonal disease.